

HORIZONTAL PROJECTION

15/11/25

Take g as 9.8 m s^{-2} unless instructed otherwise.

1. A particle is projected horizontally, at 20 m s^{-1} , from a point 78.4 m above a horizontal surface. Find:
 - a. the time taken by the particle to reach the surface
 - b. the horizontal distance travelled in that time.
2. A particle is projected horizontally with a speed of 21 m s^{-1} . Find:
 - a. the horizontal and vertical displacements of the particle, from the point of projection, $\frac{20}{7}$ seconds after projection.
 - b. how far the particle is then from the point of projection.
3. A particle is projected horizontally from a point 2.5 m above a horizontal surface. The particle hits the plane at a point which is, horizontally, 10 m from the point of projection. Find the initial speed of the particle.
4. A particle is projected horizontally at 20 m s^{-1} . Find
 - a. the horizontal and vertical components of the velocity of the particle 2 seconds after projection.
 - b. the speed and direction of motion of the particle at this time.
5. A darts player throws a dart horizontally with a speed of 14 m s^{-1} . The dart hits the board at a point 10 cm below the level at which it was released. Find the horizontal distance travelled by the dart.
6. A tennis ball is served horizontally with an initial speed of 21 m s^{-1} from a height of 2.8 m . By what distance does the ball clear a net 1 m high situated 12 m horizontally from the server?
7. The top of a vertical tower is 20 m above ground level. When a ball is thrown horizontally from the top of this tower, it first hits the ground 24 m from the base of the tower. By how much does the ball clear a vertical wall of height 13 m situated 12 m from the tower?
8. Two vertical towers stand on horizontal ground and are of heights 40 m and 30 m . A ball is thrown horizontally from the top of the higher tower with a speed of 24.5 m s^{-1} and just clears the smaller tower. Find the distance
 - a. between the two towers
 - b. between the smaller tower and the point on the ground where the ball first lands.
9. A window in a house is situated 4.9 m above the ground. When a boy throws a ball horizontally from this window with a speed of 14 m s^{-1} , the ball just clears a vertical wall situated 10 m from the house. Determine the height of the wall and how far beyond the wall the ball first hits the ground.
10. O , A and B are three points with O on level ground and A and B respectively 3.6 metres and 40 metres vertically above O . A particle is projected horizontally from B with a speed of 21 m s^{-1} and 2 seconds later a particle is projected horizontally from A with a speed of 70 m s^{-1} . Show that the two particles reach the ground at the same time and at the same distance from O and find that distance.